

Lecture 5: Lead-lag

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
break points	brytpunkter
gain	förstärkning
phase shift	fasvridning
loop gain	kretsförstärkningen
Bandwidth	Bandbredd
Resonance frequency	Resonansfrekvens
Resonance peak	Resonanstopp
Steady-state error	Stationärt fel
overshoot	översläng
(gain) crossover frequency	skärfrekvensen
phase-crossover frequency	fas-skärfrekvensen
phase margin	fasmarginalen
gain margin	amplitudmarginalen
lead compensator	lead-länk
lag compensator	lag-länk



Outline

Recap

Bode diagram

Controller design

Stability: Nyquist criterion

Servo control problem

Performance: Speed and oscillations

Requirements on the loop gain $G_o(i\omega)$

Recap: Steady-state errors

Compensation

Lead

Lag

Example: Position Control of a Car

Recap

In the previous lecture..

- ▶ Performance metrics: T_r , M , T_s ($p\%$)
- ▶ Frequency response: $y(t) = |G(i\omega)| \sin(\omega t + \arg G(i\omega))$ when $u(t) = \sin(\omega t)$ (stable systems)
- ▶ Bode diagram
- ▶ Extra:
 - Minimum-phase system = All the zeros and all the poles strictly in LHP;
Non-minimum phase system, otherwise [+Def 13.1, Ch 13.4 AH]
 - Time delay: $y(t) = u(t - T) = \sin(\omega t - \omega T)$ when $u(t) = \sin(\omega t) \Rightarrow G(s) = e^{-sT}$

Today:

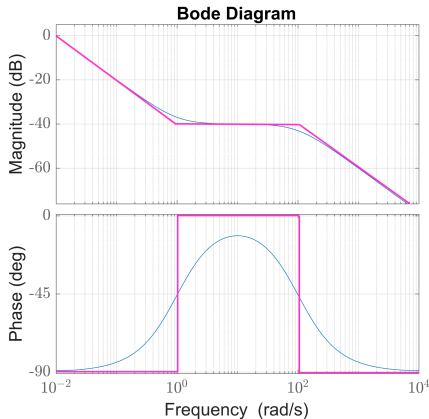
- ▶ Controller design

Recap: Bode diagram

Plot/sketch $\log |G(i\omega)|$ and $\arg G(i\omega)$ as a function of the frequency ω^* .

For the approximate Bode diagram:

- Magnitude $\log |G(i\omega)|$:
 - Find the break points and determine the slope
 - If ω is small, then slope = number of integrators
- Phase $\arg G(i\omega)$:
 - Mnemonic (informal): If the magnitude Bode plot has a line with slope $\pm k = \pm k 20 \text{ dB/decade}$, then the value of the phase is $\pm k 90^\circ$



*Remember: **Nyquist curve** = visualization of the frequency function $G(i\omega)$, where each point corresponds to a value of the angular frequency ω . **Bode diagram** = explicit illustration of dependency on ω .



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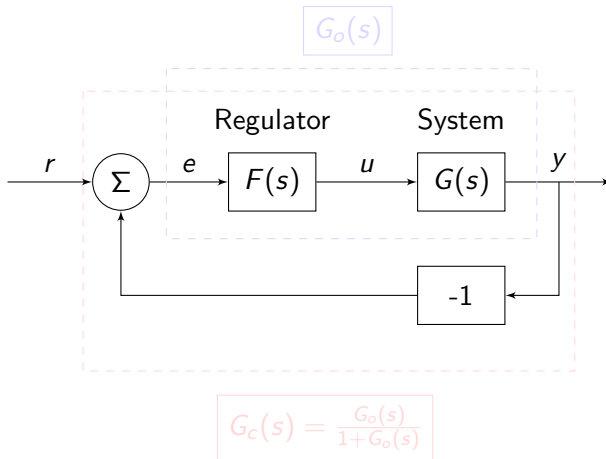
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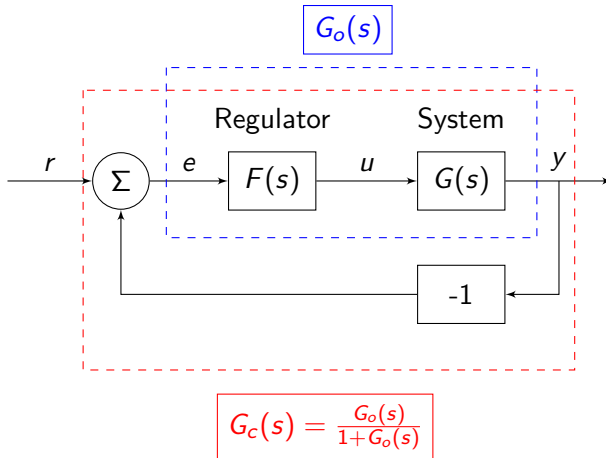
Lag

Example: Position Control of a Car

Controller design



Controller design



How is **the loop gain** $G_o(i\omega) = G(i\omega)F(i\omega)$ used to assess properties of the **closed-loop system** $G_c(s)$?

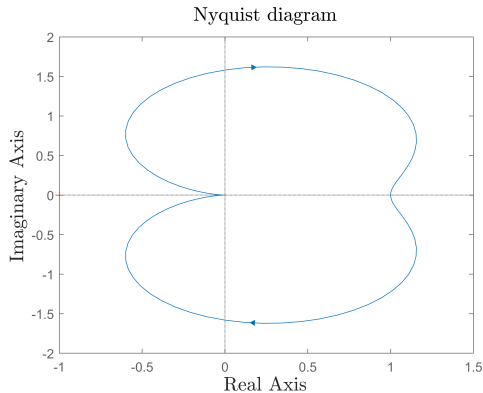
- ▶ Stability
- ▶ Speed and oscillations
- ▶ Steady-state error

Note that we can modify $G_o(i\omega)$ using the controller $F(s)$!

Stability: Nyquist criterion

Theorem (Nyquist stability criterion)

$\#$ of unstable poles of $G_o(s)$
 $=$ $\#$ of unstable poles of $G_c(s)$
 $+ \#$ of anticlockwise encirclements of -1
 made by the Nyquist plot of $G_o(s)$



Theorem (simplified Nyquist stability criterion)

Assume that $G_o(s)$ is stable (poles in $s = 0$ OK).

Then $G_c(s)$ is asymptotically stable if the Nyquist curve $G_o(i\omega)$ does not encircle -1 .

Example made with $G_o(s) = \frac{0.4}{s^2 + 0.4s + 0.4}$; poles of G_o in $-0.2 \pm 0.6 \Rightarrow$ poles of $G_c(s)$ in -0.2 ± 0.87 .

Stability: Nyquist criterion: Example

Consider $G(s) = \frac{(s+2)}{s(s-2)}$ and $F(s) = K \Rightarrow G_o(s)$ has 1 pole in RHP

$$G = (s+2)/(s*(s-2));$$

$$K = 1; F = K;$$

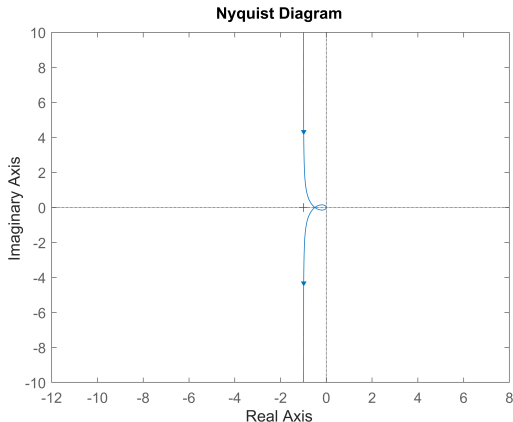
$$G_o = F*G;$$

$$\text{figure; nyquist}(G_o)$$

$$\text{ylim}([-10,10]);$$

$$\text{xlim}([-12,8])$$

To apply the Nyquist criterion,
sketch the Nyquist curve (Careful!)

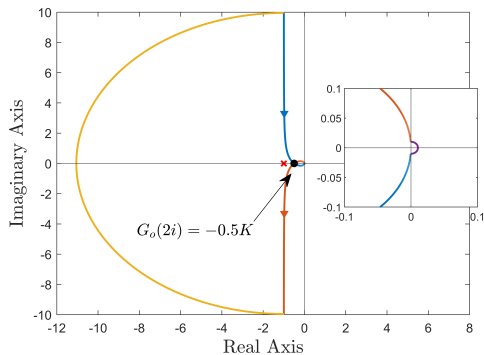


Stability: Nyquist criterion: Example

Consider $G(s) = \frac{(s+2)}{s(s-2)}$ and $F(s) = K \Rightarrow G_o(s)$ has 1 pole in RHP

$K = 1$ (unstable for $0 < K < 2$)

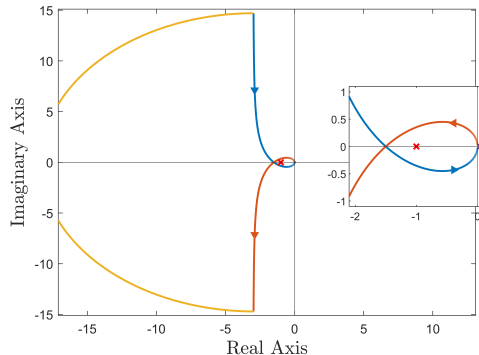
- ▶ $1 = \#$ of unstable poles of $G_c(s) + (-1)$
- ▶ $G_c(s)$ has poles in $0.5 \pm 1.3i$ at $K = 1$



— $s = i\omega, \omega > 0$
— $s = re^{i\theta}, r = 0.1, \theta \in [-\pi/2, \pi/2]$
— $s = i\omega, \omega < 0$
— $s = re^{i\theta}, R = 100, \theta \in [-\pi/2, \pi/2]$

$K = 3$ (stability for $K > 2$)

- ▶ $1 = \#$ of unstable poles of $G_c(s) + 1$
- ▶ $G_c(s)$ has poles in $-0.5 \pm 2.4i$ at $K = 3$



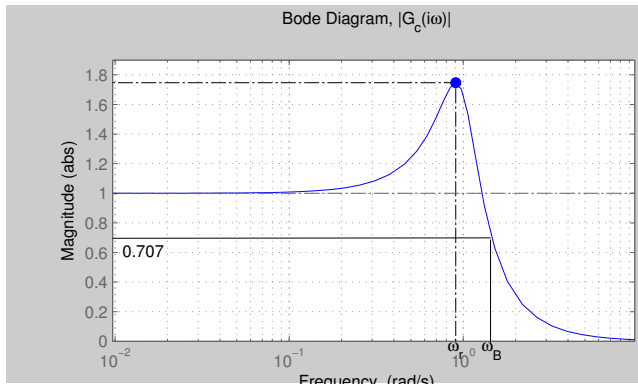
— $s = i\omega, \omega > 0$
— $s = re^{i\theta}, r = 0.2, \theta \in [-\pi/2, \pi/2]$
— $s = i\omega, \omega < 0$
— $s = re^{i\theta}, R = 100, \theta \in [-\pi/2, \pi/2]$

Servo control problem (following the reference)

The output should follow the reference signal

$$y(t) \approx r(t) \quad \Rightarrow \quad G_c(i\omega) \approx 1$$

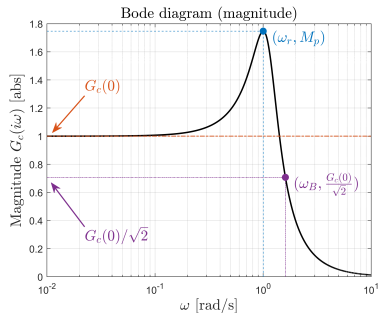
Typical appearance of $|G_c(i\omega)|$:



Performance: Speed and oscillations

From the Bode diagram we have the following quantities:

- ▶ **Bandwidth:** $|G_c(i\omega_B)| = \frac{|G_c(0)|}{\sqrt{2}}$: $\omega_B \approx$ speed
(compare with T_r)
- ▶ **Resonance frequency:** ω_r
- ▶ **Resonance peak:** $|G_c(i\omega_r)| = M_p \approx$ overshoot M
- ▶ **Steady-state error:** $e_0 = 1 - G_c(0)$



We would like

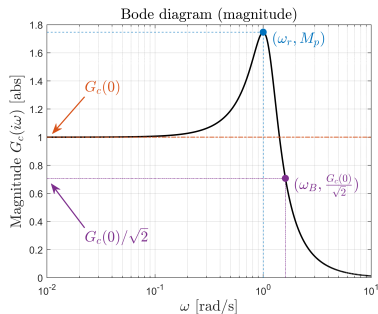
$$\begin{cases} \omega_B & \text{large} \\ M_p & \text{small} \\ e_0 & = 0 \end{cases}$$

Study Exempel 5.1 and Figur 5.8! + Connect to Exempel 3.3 and Figur 2.7

Performance: Speed and oscillations

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We would like

$$\begin{cases} \omega_B & \text{large} \\ M_p & \text{small} \\ e_0 & = 0 \end{cases}$$


$$G_c(i\omega) = \frac{G_o(i\omega)}{1 + G_o(i\omega)} \iff G_o(i\omega) = \frac{G_c(i\omega)}{1 - G_c(i\omega)}$$
$$G_c(i\omega) \approx 1 \quad \Longleftrightarrow \quad G_o(i\omega) \text{ large}$$

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Requirements on the loop gain $G_o(i\omega)$

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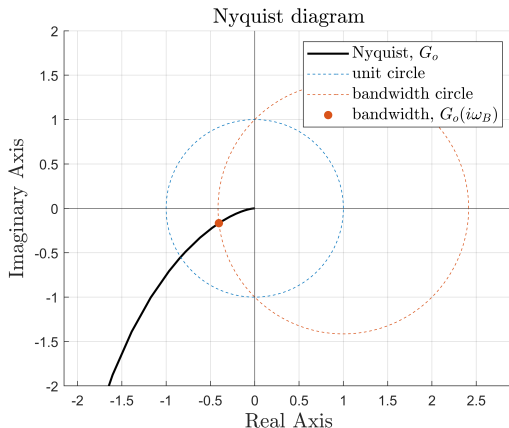
Lag

Example: Position Control of a Car

Requirements on the loop gain $G_o(i\omega)$

Bandwidth: $|G_c(i\omega)| \leq 1/\sqrt{2}$ for all $\omega \geq \omega_B$

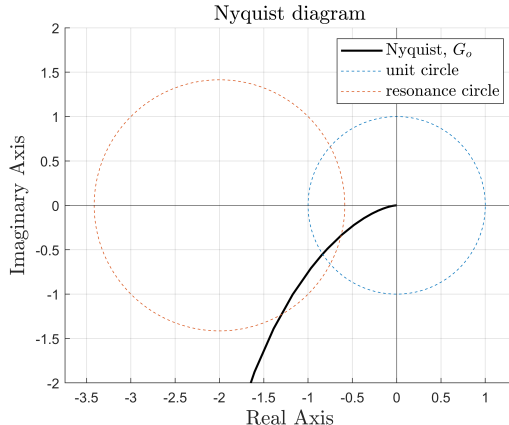
$$\Leftrightarrow |G_o(i\omega) - 1| \leq \sqrt{2} \text{ for all } \omega \geq \omega_B$$



Requirements on the loop gain $G_o(i\omega)$

Resonance peak 3dB ($= \sqrt{2}$): $|G_c(i\omega)| \leq \sqrt{2}$ for all ω

$\Leftrightarrow \sqrt{2} \leq |G_o(i\omega) + 2|$ for all ω (Not satisfied here!)





- Maps the open-loop system to the closed-loop system: $f(z) = \frac{z}{1+z}$

Proof:

$$\Rightarrow x^2 + y^2 \leq 0.5((1+x)^2 + y^2)$$

$$\Leftrightarrow (x-1)^2 + y^2 \leq 2$$

Proof: Möbius transformation

Conformal mapping that maps circles to circles

- Maps the open-loop system to the closed-loop system: $f(z) = \frac{z}{1+z}$
- Bandwidth: $\left| \frac{z}{1+z} \right| \leq \frac{1}{\sqrt{2}} \Leftrightarrow |z - 1| \leq \sqrt{2}$

Proof:

$$z = x + iy$$

$$\Rightarrow x^2 + y^2 \leq 0.5((1+x)^2 + y^2)$$

$$\Leftrightarrow x^2 - 2x + 1 + y^2 \leq 2$$

$$\Leftrightarrow (x-1)^2 + y^2 \leq 2$$

Example:

$$\text{Resonance peak} = \sqrt{2} = 3\text{dB} : \quad \left| \frac{z}{1+z} \right| \leq \sqrt{2} \Leftrightarrow \sqrt{2} \leq |z+2|$$



- Maps the open-loop system to the closed-loop system: $f(z) = \frac{z}{1+z}$
- Bandwidth: $\left| \frac{z}{1+z} \right| \leq \frac{1}{\sqrt{2}} \Leftrightarrow |z - 1| \leq \sqrt{2}$

$$z = x + iy$$

$$\Rightarrow x^2 + y^2 \leq 0.5((1+x)^2 + y^2)$$

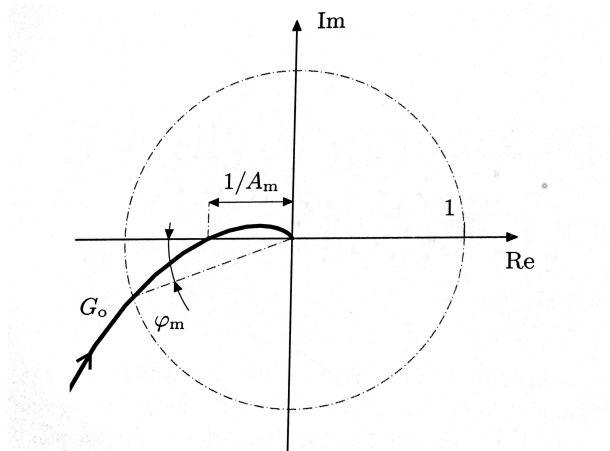
$$\Leftrightarrow x^2 - 2x + 1 + y^2 \leq 2$$

$$\Leftrightarrow (x-1)^2 + y^2 \leq 2$$

$$\text{Resonance peak} = \sqrt{2} = 3dB : \quad \left| \frac{z}{1+z} \right| \leq \sqrt{2} \quad \Leftrightarrow \quad \sqrt{2} \leq |z+2|$$

Requirements on the loop gain $G_o(i\omega)$

We will use the following quantities and corresponding frequencies from the Nyquist curve $G_o(i\omega)$ to determine speed and oscillations

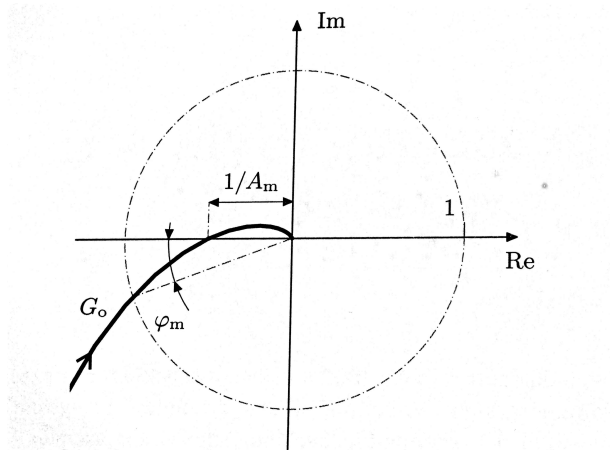


Study Figur 5.2! and Figur 5.3

The Nyquist curve $G_o(i\omega)$

- ▶ *(gain) crossover frequency / skärfrekvensen* ω_c : $|G_o(i\omega_c)| = 1$.
- ▶ *phase-crossover frequency / fas-skärfrekvensen* ω_p : $\arg(G_o(i\omega_p)) = -180^\circ$.
- ▶ *phase margin / fasmarginalen* φ_m : $\varphi_m = \arg(G_o(i\omega_c)) - (-180^\circ)$.
- ▶ *gain margin / amplitudmarginalen* A_m : $A_m = \frac{1}{|G_o(i\omega_p)|}$.

Nyquist criterion



For *well-behaved systems*, a stability requirement is

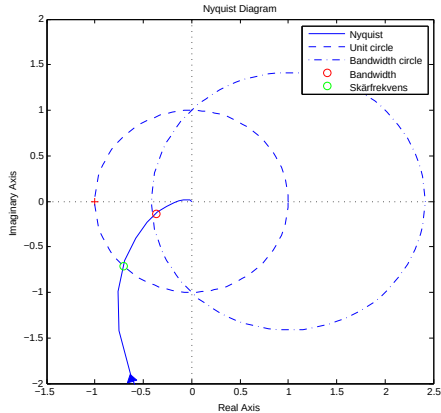
$$\begin{cases} \varphi_m > 0 \\ A_m > 1 \end{cases}$$

Relationship between bandwidth and crossover frequency

Rule of thumb: $\omega_B \approx \omega_c$

- ▶ bandwidth / bandbredd ω_B in G_c !
- ▶ crossover frequency / skärfrekvens ω_c in G_o !

Example: Here $\omega_B > \omega_c$




$$M_p \geq |G_c(i\omega_c)| = \frac{1}{2 \sin(\varphi_m/2)}$$
$$\Rightarrow G_c(i\omega_c) \approx \frac{-1}{1-1}$$

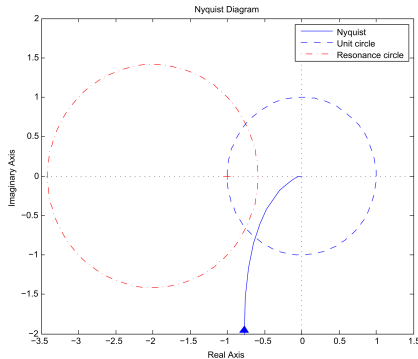
Read p. 100–103, “Att gå från G_c till G_o ”

Requirements on the phase margin

Resonance peak 3dB = $\sqrt{2}$:

$$\frac{1}{2 \sin(\varphi_m/2)} \leq \sqrt{2} \Rightarrow \varphi_m \geq 41^\circ$$

OK here!



Speed and oscillations

Requirements on speed and oscillations give requirements on ω_c , φ_m , and A_m .

Summary: It is *easier* to design the controller by modifying the loop gain

$$G_o = FG$$

rather than the closed-loop system

$$G_c = \frac{FG}{1 + FG}$$



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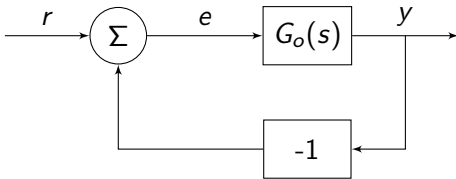
Compensation

Lead

Lag

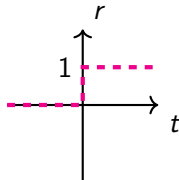
Example: Position Control of a Car

Steady-state errors



$$E(s) = \frac{1}{1+G_o(s)} R(s)$$

Let $r(t)$ be a unit step.



Then $R(s) = \frac{1}{s}$.


$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow \infty} \frac{s}{1 + G_o(s)} \cdot \frac{1}{s} = \frac{1}{1 + G_o(0)}$$
$$\lim_{t \rightarrow \infty} e(t) = 0 \text{ if } G_o(0) = \infty$$

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Example: Position Control of a Car



Determine the controller $F(s)$ so that the loop gain $G_o(s) = F(s)G(s)$ fulfills desired specifications:

- I. speed $\Rightarrow$ requirement on ω_c
- II. damping $\Rightarrow$ requirement on φ_m
- III. steady-state error $\Rightarrow$ requirement on $|G_o(0)|$

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I. Speed via P control

A P-controller is sufficient

$$F(s) = K$$

This shifts the magnitude curve (up or down), but does not change the phase curve.

Sets the crossover frequency ω_c !


$$F_{\text{lead}}(s) = \frac{\tau_D s + 1}{\beta \tau_D s + 1} \quad \beta < 1$$

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III. Steady-state error via a lag compensator

Use a PI controller ($\gamma = 0$) formulated using a *lag compensator*:

$$F_{\text{lag}}(s) = \frac{\tau_I s + 1}{\tau_I s + \gamma}$$

Fixes the steady-state error, but may worsen the phase margin!

I + II + III. via lead-lag compensator

The complete controller is given by:

$$F(s) = K \cdot F_{\text{lead}}(s) \cdot F_{\text{lag}}(s) = K \cdot \frac{\tau_D s + 1}{\beta \tau_D s + 1} \cdot \frac{\tau_I s + 1}{\tau_I s + \gamma}$$

How should we determine τ_D , τ_I , β , K , and γ ?

Lead

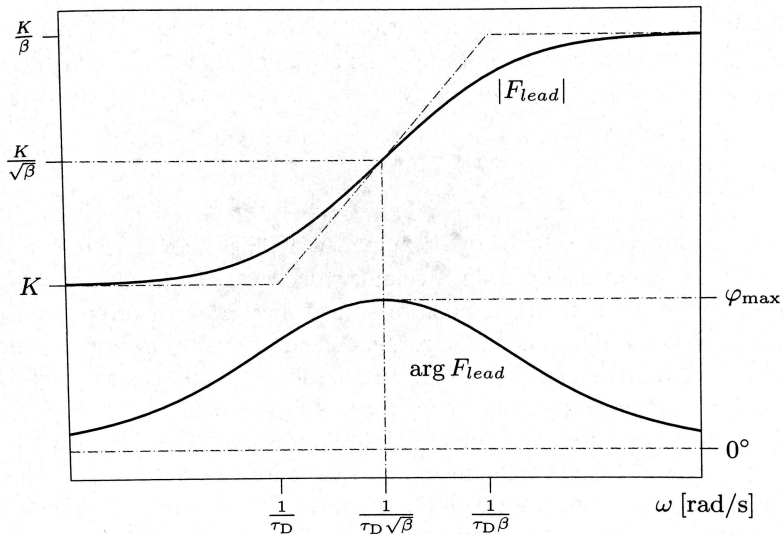
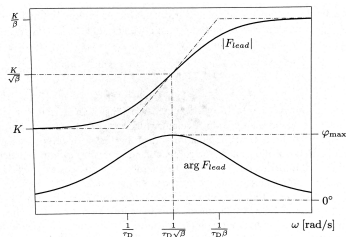


Figure 5.14 on p. 107

Lead: Some observations



- $\arg(F_{\text{lead}}(i\omega)) > 0^\circ \quad \forall \omega$ raises the phase $\Rightarrow$ Better φ_m .
- $\varphi_{\text{max}} := \max_{\omega} \arg(F_{\text{lead}}(i\omega)) = \arctan\left(\frac{1-\beta}{2\sqrt{\beta}}\right)$ obtained at $\omega = \frac{1}{\tau_D\sqrt{\beta}}$

Note that

- $\varphi_{\text{max}} \rightarrow 90^\circ$ as $\beta \rightarrow 0$
 - At this ω , we have $|F_{\text{lead}}(i\frac{1}{\tau_D\sqrt{\beta}})| = \frac{1}{\sqrt{\beta}}$
- $|F_{\text{lead}}(i\omega)| = \begin{cases} 1 & \omega \text{ small} \\ \frac{1}{\beta} & \omega \text{ large} \end{cases}$







Procedure for P controller + lead compensator

1. Determine the desired $\omega_B \Rightarrow$ Required desired crossover frequency ω_c
2. Determine the desired φ_m (usually $45^\circ - 60^\circ$) $\Rightarrow$ Required phase increase $\varphi_{\max}$ at ω_c
3. Determine β from the diagram of $\varphi_{\max}$ in Figur 5.13 (or calculate it)
4. From $\omega_c = \frac{1}{\tau_D \sqrt{\beta}}$ (see properties of lead compensator) $\Rightarrow \tau_D = \frac{1}{\sqrt{\beta} \cdot \omega_c}$
5. Choose K such that $|G(i\omega_c)| \cdot \frac{1}{\sqrt{\beta}} \cdot K = 1$, i.e., $K = \frac{\sqrt{\beta}}{|G(i\omega_c)|}$

Note! If a large phase increase is needed, use several compensators in series.

Procedure for P controller + lead compensator

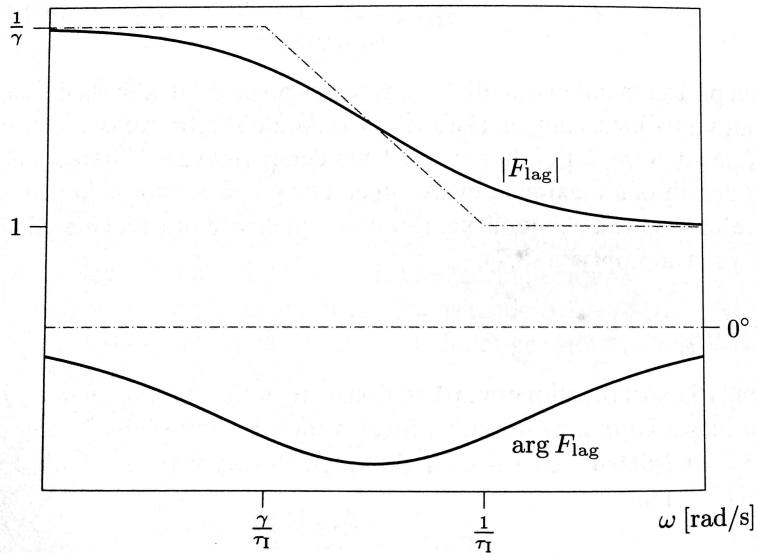
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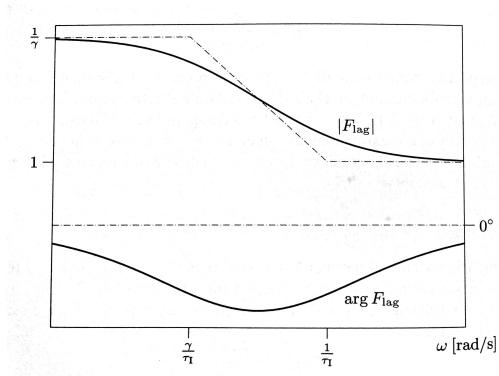
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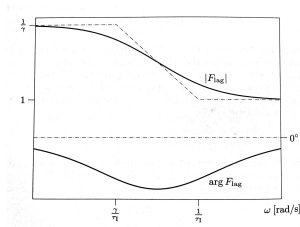
III. Steady-state error via PI / lag

$$F_{\text{lag}}(s) = \frac{\tau_I s + 1}{\tau_I s + \gamma}$$



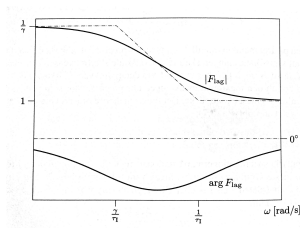
- ▶ Swedish: Fasretarderande länk; reduces to PI when $\gamma = 0$
- ▶ $F_{\text{lag}}(0) = \frac{1}{\gamma} \Rightarrow \gamma \text{ small} \Rightarrow |G_o(0)| \text{ large} \Rightarrow \text{small steady-state error}$
- ▶ $\arg F_{\text{lag}}(i\omega) < 0 \quad \forall \omega \text{ worsens } \varphi_m \xrightarrow{\gamma=0} \arg F_{PI}(i\omega_c) = -\arctan \frac{1}{\tau_I \omega_c}$

Procedure for the PI/lag compensator



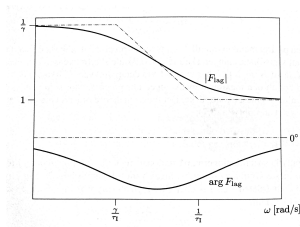
- Specify $e_0 \Rightarrow$ Calculate γ
- Try (rule of thumb) $\tau_I = \frac{10}{\omega_c} \Rightarrow \arg F_{\text{lag}}(i\omega_c) \geq -5.7^\circ$, i.e., small effect on φ_m . Add an additional 5.7° when you consider the phase boost at step 2 of the F_{lead} procedure
- If the error decreases too slowly $\Rightarrow$ decrease τ_I and compensate for any deterioration of φ_m with F_{lead}

Procedure for the PI/lag compensator



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- Try (rule of thumb) $\tau_I = \frac{10}{\omega_c} \Rightarrow \arg F_{\text{lag}}(i\omega_c) \geq -5.7^\circ$, i.e., small effect on φ_m .
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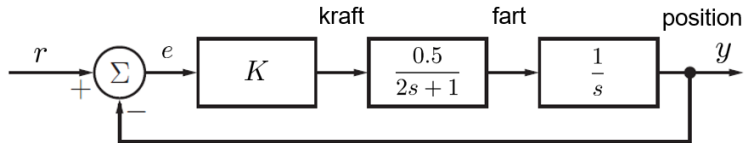
Compensation

Easier to understand by working through examples!

Lots of practice through exercises and the project assignment (Lab 3)

Example: Position Control of a Car

Iterative design of a compensator! First **I. twice as fast** and then **II. preserve oscillatory behavior/damping**.

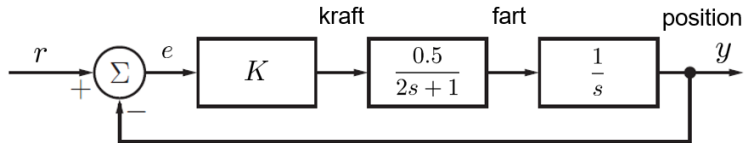


- ▶ $G(s) = \frac{0.5}{s(2s + 1)}$
- ▶ Starting point: $K = 1$ gives $T_r = 3.28$ s, $\omega_B = 0.636$ rad/s, and $\omega_c = 0.393$ rad/s.
- ▶ **I. Twice as fast:** Calculate $F(s) = K$ such that $\omega_c = 2 \cdot 0.393 = 0.786$ rad/s.

$$1 = |G_o(i\omega_c)| = K \left| \frac{0.5}{(2i\omega_c + 1)i\omega_c} \right| = 0.34K \Rightarrow \boxed{K = 2.94}$$

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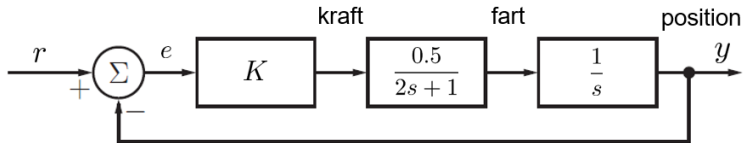


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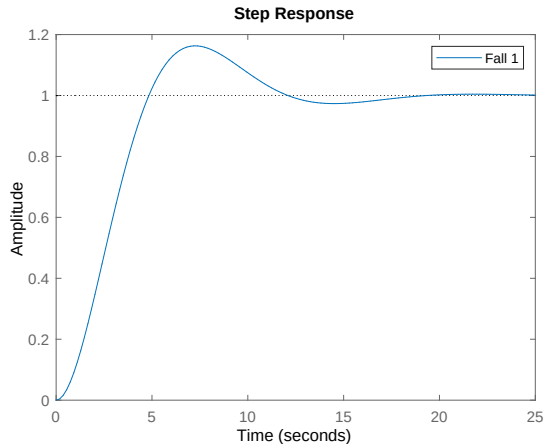
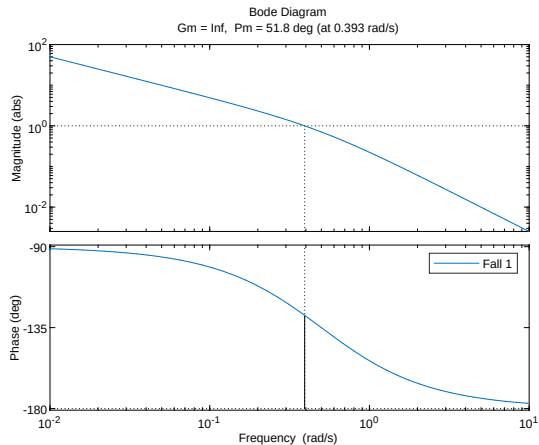


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Before compensation:

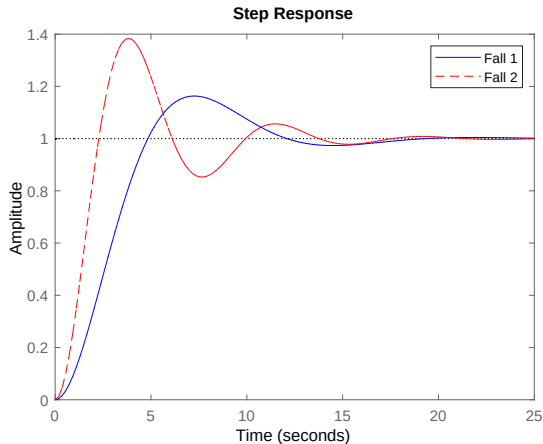
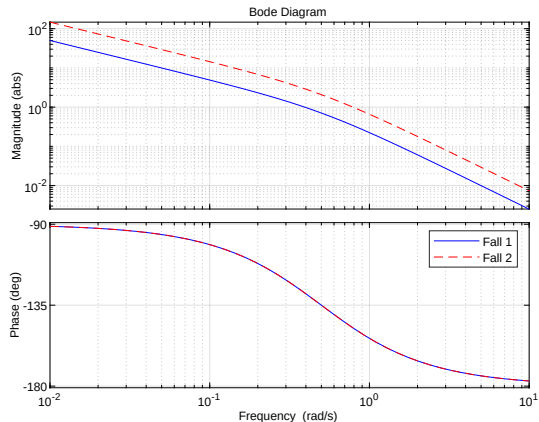


Remember: ω_c is such that $|G_o(i\omega_c)| = 1$, and $\varphi_m = \arg(G_o(i\omega_c)) - (-180^\circ)$

Homework: By looking at the Bode plot, derive $\varphi_m \approx 52^\circ$

Example: Position Control of a Car

After compensation with $F(s) = K = 2.94$ (P control):



- Performance: $T_r = 1.53$ s, $\omega_B = 1.25$ rad/s, and $\omega_c = 0.788$ rad/s (cf. starting point)
- Speed requirement satisfied, but oscillations increased (from $M = 16\%$ to $M = 38\%$)!
- Homework: By looking at the Bode plot, derive $\varphi_m \approx 32^\circ$ (before it was 52° !)

Example: Position Control of a Car

II. Preserve oscillatory behavior/damping

- Phase margin $\varphi_m = 52^\circ$ before and $\varphi_m = 32^\circ$ after compensation with $K = 2.94$
- Increase the phase at the new desired crossover frequency $\omega_c = 0.786$ rad/s by $52^\circ - 32^\circ = 20^\circ$. Use phase lead compensation with $F_{\text{lead}}(s) = \frac{\tau_D s + 1}{\beta \tau_D s + 1}$!

- First solve $\arctan\left(\frac{1}{2} \frac{1-\beta}{\sqrt{\beta}}\right) = 20^\circ \Rightarrow \boxed{\beta = 0.49}$ (numerically or using Fig. 5.13)
- Make sure that the phase boost occurs at the desired crossover frequency:

$$\omega_c = \frac{1}{\tau_D \sqrt{\beta}} \Rightarrow \tau_D = \frac{1}{\sqrt{\beta} \cdot \omega_c} = \frac{1}{\sqrt{0.49} \cdot 0.786} \Rightarrow \boxed{\tau_D = 1.81 \text{ s}}$$

- Readjust K so that the crossover frequency is preserved:

$$1 = |G_0(i\omega_c) = K|F_{\text{lead}}(i\omega_c)| \left| \frac{0.5}{(2i\omega_c + 1)i\omega_c} \right| = 0.34K/\sqrt{\beta} \Rightarrow \boxed{K = 2.06}$$

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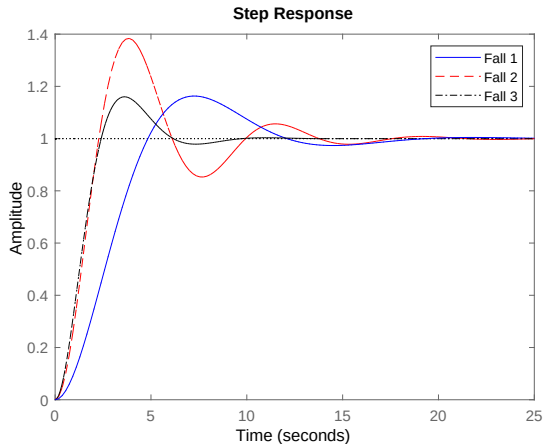
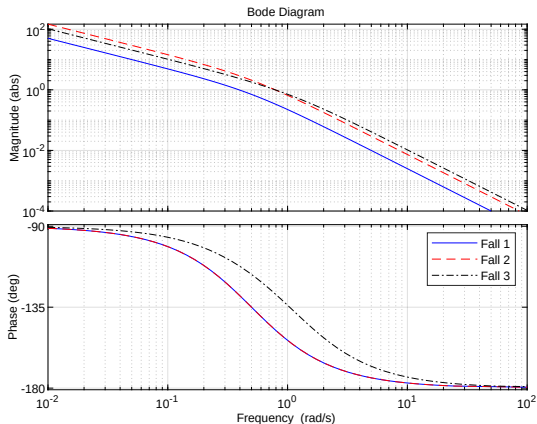
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After compensation with $F(s) = KF_{\text{lead}}(s)$:



- Performance: $T_r = 1.63$ s, $\omega_B = 1.27$ rad/s, $\omega_c = 0.787$ rad/s, and $M = 16\%$ (cf. starting point)
- The requirements for speed and oscillatory behavior/damping are satisfied!